MILITARY SPECIFICATION

RADIACMETER, IM-239/WDQ

This amendment forms a part of Military Specification MIL-R-28881(EC), dated 26 March 1984, and is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1.1, line 2: Delete "the" before "listed".

2.1.1, "SPECIFICATIONS", "MILITARY": Delete "MIL-R-28760 Radiacmeter, Air Particle Detector, IM-192/WDQ".

2.1.1, "STANDARDS", "MILITARY": Delete "MIL-STD-810 Environmental Test Methods And Engineering Guidelines" and substitute "MIL-STD-810C Environmental Test Methods".

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3.2.1, line 2: Add "dimensional" before "environment".

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* FIGURE 1: Delete and substitute the attached new FIGURE 1.

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FIGURE 2: Delete "17.0" and substitute "17.00 ±0.10".

FIGURE 2: Delete "2 3/32" and substitute "23/32".

3.3.1.1, line 3: Delete "end points beta" and substitute "beta end point energies".

3.3.1.1, lines 3 and 4: Delete "megaelectronvolts" and substitute "million electron-volt".

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FIGURE 3: Delete "17.00 ±.031" and substitute "17.00 ±0.10".

3.3.1.5, after the last sentence, add: "The radiacmeter may start up with a preset alarm value of 1×10^{-9}.".

3.3.1.6: Delete the text and substitute: "The range of detection shall be linear such that the total system error shall not exceed ±10 percent from actual radioactivity from 2×10^{-11} to 1×10^{-7} µCi/ml and ±15 percent in the last range from 1×10^{-7} to 1×10^{-6} µCi/ml. Any reading within the five decades of range shall be capable of being read on the front console.".

3.3.1.8, lines 2 and 3: Delete "shall be equal to or greater than" and substitute "may be equal to".

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3.3.1.9, line 5: Delete "such a" and substitute "such as".

3.3.2.3: Delete the text and substitute: "The radiacmeter shall fit the mounting holes as dimensioned in FIGURE 2. The air inlet and outlet ports shall use 7/8-14 UNF-28 fittings as specified in MS16142.".

3.3.4.2.1a, line 3: Delete "bave" and substitute "have".

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3.3.4.2.1c, line 2: Delete "air regulation and cut-off" and substitute "paper feed".

3.3.4.2.3b, line 4: After "test jacks" add "or connectors".

3.3.5.2, lines 2 and 3: Delete "0°C to 60°C" and substitute "5°C to 50°C".

3.3.5.2, line 3: Delete "4.5.16.2" and substitute "4.5.16.2.2".

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3.3.5.3.2, line 3: Delete "be designed to".

3.3.5.3.2, third sentence: Delete and substitute "The sealed sampling system shall contain the steam pressurization transient specified in 3.3.5.3.4.".

3.3.5.3.2, last sentence: Delete.

3.4, line 1: Delete "3.4" and substitute "3.4.1".

3.4.1, line 2: Delete "3.4.1" and substitute "3.4.1.1".

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3.4.7.6, second sentence: Delete.

3.4.7.6, line 3: Delete "may be" and substitute "shall be".

3.4.7.7, line 2: Add "radiacmeter's" before "power supply".

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3.4.9d: Add "including front panel".

3.4.9.1, line 2: Delete "is filtered through. " and substitute "passes through the filter paper.".

3.4.9.1, line 4: Delete "positon" and substitute "position".

3.4.9.1, line 5: Delete "24 hours" and substitute "8 hours".

3.4.9.1, after the last sentence, add: "A minimum of 75 feet of filter paper will be provided.".

3.4.9.3: Add "The preamplifier assembly shall receive the electrical pulses from the photomultiplier tube and process and amplify each pulse so that the pulses can be counted by the microprocessor. Discrimination against electrical noise pulses shall be provided.".

3.4.9.4, title: Delete and substitute "Input power and control assembly, including front panel."

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3.4.9.8, line 3: Delete "and volume" and substitute "air volume".

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TABLE I, "Low temperature" and "High temperature", Requirement paragraph: Delete "3.3.5.2" and substitute "3.3.5.1".

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4.4.1, line 2: Delete the period after "delivery" and add ", except as specified under "Sample size" column in TABLE II.".

TABLE II, "Weight": Delete entry in its entirety.

TABLE II, "Safety": Delete "3.4.7" and substitute "3.4.7.1, 3.4.7.10".

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TABLE II, "Calibration": Indent one space under "Performance".

TABLE II, "Pressure": Delete "3.3.5.3" and substitute "3.3.5.3.2"; delete "4.5.17" and substitute "4.5.17.1".

TABLE II: Change "Sample size" from "2" to "2 per contract" for the following:

Airborne and structureborne noise
Steady-state voltage and frequency
Power interruption
Power and power factor

TABLE II: Delete "EMI" and substitute "EMC".

4.5: Delete "4.5.18.1" and substitute "4.5.18".

4.5.1, line 3: Delete "4.5.18.1" and substitute "4.5.18".

4.5.2, lines 1, 9, 10, and 11: Delete "Sr-90 and Y-90" and substitute "Sr-90-Y-90".

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4.5.3, line 9: Delete "µCi/cu cm" and substitute "µCi/ml".

4.5.8, line 5: Delete "at 75 percent" and substitute "at least 75 percent".

4.5.10, line 3: Delete "less that" and substitute "less than".

4.5.10, lines 4 and 6: Delete "check" and substitute "test".

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4.5.16: Delete "and" and substitute "through".

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FIGURE 6, Upper operating temperature: Delete dotted line to 60°C and substitute dotted line to 50°C.

FIGURE 6, Lower operating temperature: Delete "0°C" and substitute "5°C".

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4.5.17.1: Add "Pressure relief valve shall be blocked or kept from operating during this test."

4.5.17.3, line 1: Delete "100°F" and substitute "38°C (100°F)".

4.5.17.3, line 2: Delete "235°F" and substitute "113°C (235°F)".

4.5.17.3, line 4: Delete "approximately 20 minutes" and substitute "20 to 30 minutes".

After "4.5.17.3", add the following new paragraph:

"4.5.17.4 Relief valve test. The radiacmeter shall be pressurized above 70 psig and the pressure relief valve shall actuate between 70 psig and 80 psig.".

4.5.18: Delete and renumber "4.5.18.1" to "4.5.18".

4.6, line 2: Delete "Test Plan XXC" and substitute "Test Plan 30-5".

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6.3, second line: Delete "7-104.55 of the Defense Acquisition Regulations." and substitute "52.209-3 of the Federal Acquisition Regulations.".

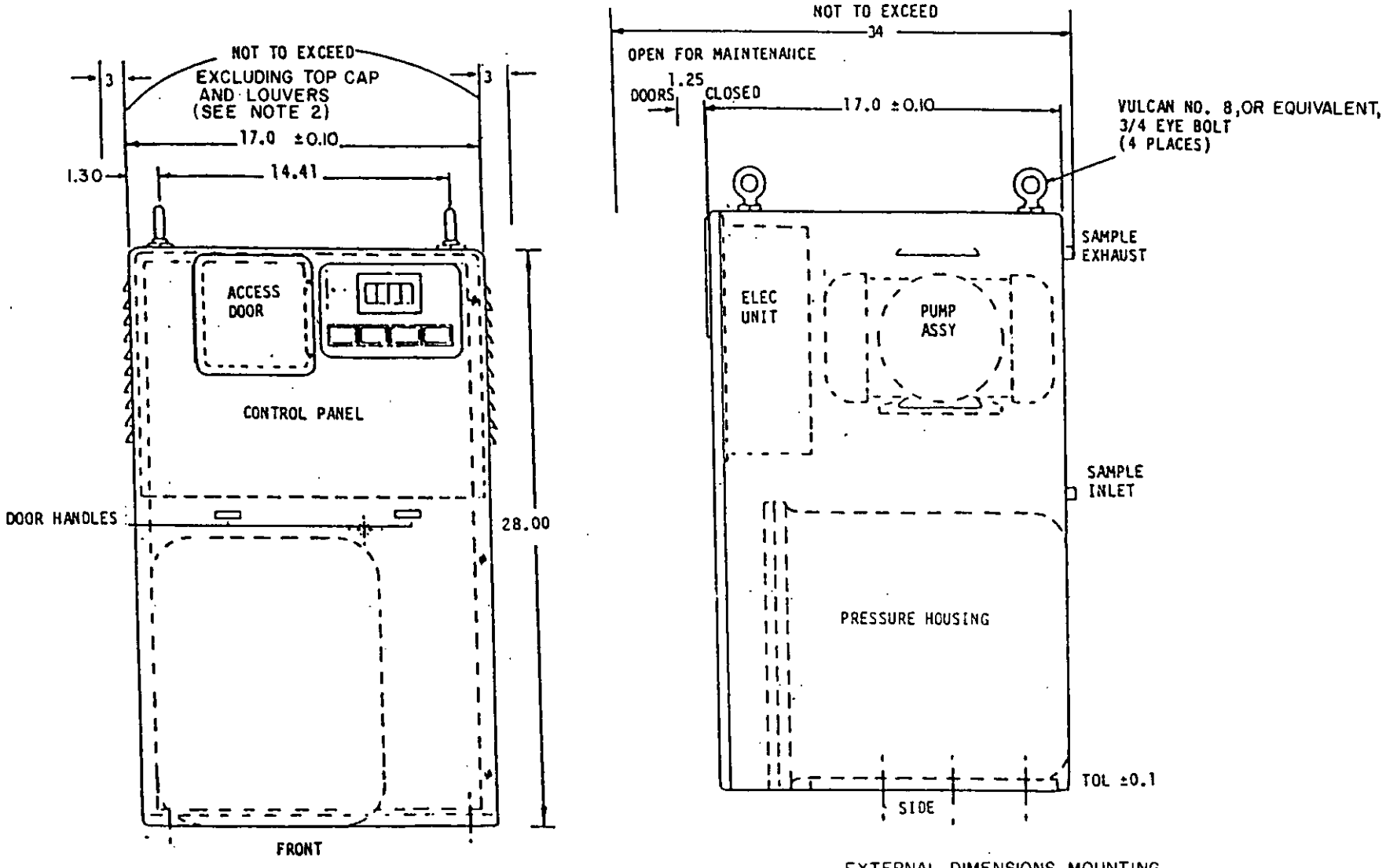
6.3, after the last sentence, add: "The first article inspection will normally be required if the prospective contractor has no previous experience constructing the radiacmeter.".

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NOTE: 1. Dimensions are in inches.
2. Louvers, top cap, handles and bails are not included in 17.0 X 17.0 dimensions.

IM-239 XN-1
RADIACMETER

EXTERNAL DIMENSIONS, MOUNTING, PATTERN I/O AND INPUT POWER LOCATIONS SAME AS EXISTING RADIACMETER, IM-192 WDQ

FIGURE 1. Radiacmeter, IM-239 XN-1/WDQ, external dimensions and maximum envelope.

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